

DIGITAL SAT MATH MASTERY SERIES

DATA INTERPRETATION FROM TABLES AND GRAPHS

Total Questions: 50 | Easy: 8 | Medium: 19 | Hard: 23

DOCUMENT 2 — SOLUTIONS & EXPLANATIONS

Aligned to Digital SAT Question Style & Cognitive Demand

All questions original — fresh numbers, fresh contexts

Topic	Data Interpretation from Tables and Graphs
Document	DOCUMENT 2 — SOLUTIONS & EXPLANATIONS
Questions	50
Difficulty	Easy: 8 Medium: 19 Hard: 23
Format	Multiple Choice (4 options)
Exam Target	Digital SAT (DSAT)

DATA INTERPRETATION FROM TABLES AND GRAPHS — FULL SOLUTIONS & EXPLANATIONS

Q1. [EASY]

✓ Correct Answer: A) 595

The table below shows the number of visitors to a museum each day for one week. Day | Mon | Tue | Wed | Thu | Fri | Sat | Sun Visitors | 142 | 98 | ...

STEP-BY-STEP SOLUTION:

Identify the two highest days: Saturday (320) and Sunday (275). Total: $320 + 275 = 595$.

WHY EACH OPTION IS WRONG:

✓ A) CORRECT: $320 + 275 = 595$.

✗ B) $508 = 320 + 188$ (uses Friday instead of Sunday).

✗ C) 623 is not a valid sum of any two days.

✗ D) $520 = 275 + 245$, which doesn't match any day.

TRAP ALERT ■

Including Friday (188) instead of Sunday (275) as the second highest day.

KEY INSIGHT ■

Rank the days first before summing: Sat 320, Sun 275, Fri 188.

FAST METHOD ■

List and sort: 320, 275, 201, 188, 142, 115, 98. Top 2 = $320 + 275 = 595$.

Q2. [EASY]

✓ Correct Answer: D) Ethan

The bar graph below represents the number of books read by five students during summer. Student | Books Read Ana | 8 Ben | 5 Carlos | 12 Diana ...

STEP-BY-STEP SOLUTION:

Ben read 5 books. Twice as many = 10. Ethan read 10. Answer: Ethan.

WHY EACH OPTION IS WRONG:

✗ A) Ana read 8 books. $8 \neq 2 \times 5 = 10$.

✗ B) Carlos read 12. $12 \neq 10$.

✗ C) Diana read 7. $7 \neq 10$.

✓ D) CORRECT: Ethan read $10 = 2 \times 5$.

TRAP ALERT ■

Computing $5 \times 2 = 10$ correctly but then matching 10 to the wrong student.

KEY INSIGHT ■

Double-check each student's count against your computed value (10).

FAST METHOD ■

$2 \times \text{Ben} = 2 \times 5 = 10$. Find who has 10: Ethan ✓

Q3. [EASY]**✓ Correct Answer: A) 3.0 inches**

The table shows the monthly rainfall (in inches) for a city over four months. Month | Jan | Feb | Mar | Apr
Rainfall | 3.2 | 2.8 | 4.1 | 1.9 What is ...

STEP-BY-STEP SOLUTION:

Mean = $(3.2 + 2.8 + 4.1 + 1.9)/4 = 12.0/4 = 3.0$ inches.

WHY EACH OPTION IS WRONG:

✓ **A)** CORRECT: $12.0 \div 4 = 3.0$.

✗ **B)** 2.95 would result from summing to 11.8.

✗ **C)** 3.25 would result from summing to 13.0.

✗ **D)** 2.85 would result from summing to 11.4.

TRAP ALERT ■

Adding incorrectly: e.g., $3.2 + 2.8 = 6.0$, then $6.0 + 4.1 = 9.1$ (forgetting the +1 carry), etc.

KEY INSIGHT ■

Sum first carefully, then divide. $3.2 + 2.8 = 6.0$; $6.0 + 4.1 = 10.1$; $10.1 + 1.9 = 12.0$.

FAST METHOD ■

$12.0 \div 4 = 3.0$ ✓

Q4. [EASY]**✓ Correct Answer: B) 5**

A scatterplot shows the relationship between hours studied (x) and test score (y). The line of best fit passes through the points (2, 65) and (8, 95). What is t ...

STEP-BY-STEP SOLUTION:

Slope = $(95 - 65)/(8 - 2) = 30/6 = 5$.

WHY EACH OPTION IS WRONG:

✗ **A)** 4 would come from $30/7.5$ or a similar error.

✓ **B)** CORRECT: slope = $30/6 = 5$.

✗ **C)** 6 comes from dividing 30 by 5.

✗ **D)** 3 comes from incorrect subtraction in numerator or denominator.

TRAP ALERT ■

Inverting rise and run: computing $(8 - 2)/(95 - 65) = 6/30 = 0.2$.

KEY INSIGHT ■

Slope = $(y_{\blacksquare} - y_{\blacksquare})/(x_{\blacksquare} - x_{\blacksquare})$. y goes in numerator.

FAST METHOD ■

$(95 - 65)/(8 - 2) = 30/6 = 5$ ✓

Q5. [EASY]**✓ Correct Answer: B) \$0.60**

The table below shows the price of gasoline at a station over five weeks. Week | 1 | 2 | 3 | 4 | 5 Price | 3.20 | 3.35 | 3.50 | 3.65 | 3.80 By ho...

STEP-BY-STEP SOLUTION:

Change = $3.80 - 3.20 = 0.60$ dollars.

WHY EACH OPTION IS WRONG:

✗ **A)** \$0.50 is the change for 4 weeks at \$0.15 each — wrong count or rate.

✓ **B)** CORRECT: $3.80 - 3.20 = \$0.60$.

✗ **C)** \$0.65 likely comes from misreading a price.

✗ **D)** \$0.45 is three increments of \$0.15.

TRAP ALERT ■

Subtracting Week 2 from Week 5 instead of Week 1 from Week 5.

KEY INSIGHT ■

'From Week 1 to Week 5' = final minus initial = $3.80 - 3.20$.

FAST METHOD ■

Each week increases by \$0.15. Over 4 intervals: $4 \times \$0.15 = \0.60 ✓

Q6. [EASY]**✓ Correct Answer: B) 9 hours**

A pie chart shows how a student spends 24 hours in a day: Sleep 37.5%, School 25%, Homework 12.5%, Recreation 16.7%, Other 8.3%. How many hours per day does the...

STEP-BY-STEP SOLUTION:

37.5% of 24 hours = $0.375 \times 24 = 9$ hours.

WHY EACH OPTION IS WRONG:

✗ **A)** 8 hours = 33.3% of 24.

✓ **B)** CORRECT: $0.375 \times 24 = 9$ hours.

✗ **C)** 10 hours = 41.7% of 24.

✗ **D)** 7 hours = 29.2% of 24.

TRAP ALERT ■

Rounding 37.5% to 38% and getting $9.12 \approx 9$, or confusing 37.5% with $1/3$ (=33.3%).

KEY INSIGHT ■

$37.5\% = 3/8$. $3/8 \times 24 = 9$ exactly.

FAST METHOD ■

$3/8 \times 24 = 72/8 = 9$ ✓

Q7. [EASY]**✓ Correct Answer: D) Dune**

The table below shows the population of four towns. Town | Alton | Bexley | Croft | Dune Population | 4,820 | 7,310 | 3,095 | 5,640 Which town has the ...

STEP-BY-STEP SOLUTION:

Rank: Bexley (7,310), Dune (5,640), Alton (4,820), Croft (3,095). Second largest = Dune.

WHY EACH OPTION IS WRONG:

- ✗ **A)** Alton is 3rd largest.
- ✗ **B)** Bexley is 1st (largest).
- ✗ **C)** Croft is 4th (smallest).
- ✓ **D)** CORRECT: Dune is 2nd largest at 5,640.

TRAP ALERT ■ Choosing the largest instead of the second-largest.

KEY INSIGHT ■ Re-read: 'second-largest.' Rank all values first.

FAST METHOD ■ Sorted: 7310 > 5640 > 4820 > 3095. Second = 5,640 = Dune ✓

Q8. [EASY]**✓ Correct Answer: C) Q3 to Q4**

A line graph shows a company's quarterly profits (in thousands of dollars): Q1: \$45k, Q2: \$52k, Q3: \$38k, Q4: \$61k. Between which two consecutive quarters did p...

STEP-BY-STEP SOLUTION:

Q1→Q2: +7. Q2→Q3: -14 (decrease). Q3→Q4: +23. Largest increase = Q3 to Q4.

WHY EACH OPTION IS WRONG:

- ✗ **A)** Q1 to Q2: increase of \$7k.
- ✗ **B)** Q2 to Q3: this is a DECREASE of \$14k.
- ✓ **C)** CORRECT: Q3 to Q4: increase of \$23k (the largest).
- ✗ **D)** Q1 to Q4 spans multiple quarters; not consecutive.

TRAP ALERT ■ Including Q2→Q3 (a decrease) when looking for 'increase.'

KEY INSIGHT ■ Only consider increases. Q3→Q4 = 61 - 38 = +23k, the largest positive jump.

FAST METHOD ■ +7, -14, +23. Largest increase = +23 (Q3 to Q4) ✓

Q9. [MEDIUM]**✓ Correct Answer: B) 12 cm**

The table below shows data on plant growth. Days of sunlight per week | 1 | 2 | 3 | 4 | 5 | 6 | 7 Height after 4 weeks (cm) | 4 | 7 | 10 | 14 | 18 | 21...

STEP-BY-STEP SOLUTION:

From the data: at 3 days → 10 cm, at 4 days → 14 cm. The difference is 4 cm over 1 day. At 3.5 days (halfway between 3 and 4): height $\approx 10 + 0.5 \times 4 = 12$ cm.

WHY EACH OPTION IS WRONG:

- ✗ **A)** 11 cm would be the value at roughly 3.25 days.
- ✓ **B)** CORRECT: linear interpolation gives 12 cm at 3.5 days.
- ✗ **C)** 13 cm would be at 3.75 days.
- ✗ **D)** 10 cm is the value at exactly 3 days.

TRAP ALERT ■ Reading 3.5 days as 3 days and giving the answer for day 3.

KEY INSIGHT ■ For non-tabulated values, interpolate (find midpoint between surrounding data points).

FAST METHOD ■ 3 days → 10, 4 days → 14. Midpoint (3.5 days) → $(10+14)/2 = 12$ ✓

Q10. [MEDIUM]**✓ Correct Answer: B) 44%**

The two-way table below shows survey results from 200 students about their preferred study location. | Library | Home | Café | Total Freshmen ...

STEP-BY-STEP SOLUTION:

Freshmen in library = 24. Total in library = 54. Percentage = $24/54 \times 100 = 44.4\% \approx 44\%$.

WHY EACH OPTION IS WRONG:

- ✗ **A)** 32% would use 24/75 (freshmen in library out of all freshmen).
- ✓ **B)** CORRECT: $24/54 \approx 44\%$.
- ✗ **C)** 45% is close but slightly off — wrong denominator.
- ✗ **D)** 48% likely uses 24/50 or similar error.

TRAP ALERT ■ Dividing 24 by 75 (all freshmen) instead of 54 (all library users).

KEY INSIGHT ■ The question asks for '% of library users who are freshmen' → denominator = total library users = 54.

FAST METHOD ■ $24/54 = 0.444... \approx 44\%$ ✓

Q11. [MEDIUM]**✓ Correct Answer: B) 15 mpg**

The scatterplot below displays data for 10 cars, showing engine size (liters) on the x-axis and fuel efficiency (mpg) on the y-axis. The line of best fit has eq...

STEP-BY-STEP SOLUTION:

$$y = -6(4.5) + 42 = -27 + 42 = 15 \text{ mpg.}$$

WHY EACH OPTION IS WRONG:

✗ **A)** 12 mpg: $-6(5) + 42 = 12$. This uses engine size 5, not 4.5.

✓ **B)** CORRECT: $-6(4.5) + 42 = 15$.

✗ **C)** 18 mpg: $-6(4) + 42 = 18$. Uses engine size 4.

✗ **D)** 9 mpg: $-6(5.5) + 42 = 9$. Wrong engine size.

TRAP ALERT ■ Plugging in 4 or 5 instead of 4.5.

KEY INSIGHT ■ Substitute $x = 4.5$ exactly into the equation.

FAST METHOD ■ $-6 \times 4.5 = -27$. $-27 + 42 = 15$ ✓

Q12. [MEDIUM]**✓ Correct Answer: A) $H = -6T + 210$**

A table shows temperature ($^{\circ}\text{C}$) and the number of hot drinks sold at a café. Temperature ($^{\circ}\text{C}$) | 5 | 10 | 15 | 20 | 25 Hot drinks sold | 180 | 150 | 120 | ...

STEP-BY-STEP SOLUTION:

Check: At $T = 5$, $H = -30 + 210 = 180$ ✓. At $T = 25$, $H = -150 + 210 = 60$ ✓. Slope = $(60 - 180)/(25 - 5) = -120/20 = -6$. y-intercept: $180 = -6(5) + b \rightarrow b = 210$.

WHY EACH OPTION IS WRONG:

✓ **A)** CORRECT: verified with two points.

✗ **B)** Positive slope — hot drinks would increase with temperature. Wrong.

✗ **C)** $H = -6T + 150$: at $T = 5$, $H = -30 + 150 = 120 \neq 180$.

✗ **D)** $H = -30T + 210$: at $T = 5$, $H = -150 + 210 = 60 \neq 180$.

TRAP ALERT ■ Confusing slope and intercept values; or using -30 (the difference between successive T values) as the slope.

KEY INSIGHT ■ Slope = $\Delta H / \Delta T$. Check any point to verify intercept.

FAST METHOD ■ Slope = $(150 - 180)/(10 - 5) = -30/5 = -6$. Intercept: $180 = -6(5) + b \rightarrow b = 210$.

Q13. [MEDIUM]**✓ Correct Answer: B) 81**

The table below shows the scores of 8 students on a math quiz. Student | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 Score | 72 | 85 | 91 | 68 | 77 | 95 | 83 | 79 ...

STEP-BY-STEP SOLUTION:

Sort scores: 68, 72, 77, 79, 83, 85, 91, 95. With 8 values, median = average of 4th and 5th = $(79 + 83)/2 = 162/2 = 81$.

WHY EACH OPTION IS WRONG:

✗ **A)** 82 = wrong average: $(79 + 85)/2 = 82$ (skipped a value).

✓ **B)** CORRECT: $(79 + 83)/2 = 81$.

✗ **C)** 80 = possible arithmetic error.

✗ **D)** 83 = the 5th value alone (didn't average with 4th).

TRAP ALERT ■

Forgetting to sort the data before finding the median, or only taking one of the two middle values.

KEY INSIGHT ■

Always SORT first. With even n , median = mean of $(n/2)$ th and $(n/2 + 1)$ th values.

FAST METHOD ■

Sorted: 68, 72, 77, [79, 83], 85, 91, 95. Median = $(79+83)/2 = 81$ ✓

Q14. [MEDIUM]**✓ Correct Answer: C) The income level at which the model predicts 0 years of education**

A researcher plotted income (x , in thousands of dollars) vs. years of education (y). The line of best fit is $y = 0.4x - 2$. What does the x -intercept of this lin...

STEP-BY-STEP SOLUTION:

x -intercept: set $y = 0$. $0 = 0.4x - 2 \rightarrow x = 5$. This means when income = \$5,000, the model predicts 0 years of education.

WHY EACH OPTION IS WRONG:

✗ **A)** Income at which education is zero — this is actually reworded the same way as C. But A doesn't specify the model context.

✗ **B)** Education when income is zero = y -intercept, not x -intercept.

✓ **C)** CORRECT: x -intercept = value of x when $y = 0$.

✗ **D)** Slope describes the relationship rate, not the intercept.

TRAP ALERT ■

Confusing which variable is x and which is y when describing intercepts.

KEY INSIGHT ■

x -intercept $\rightarrow$ set $y = 0 \rightarrow$ solve for x . The meaning is: what INPUT gives OUTPUT of zero?

FAST METHOD ■

$0 = 0.4x - 2 \rightarrow x = 5$. At \$5,000 income, model predicts 0 years of education.

Q15. [MEDIUM]**✓ Correct Answer: A) 7:48**

The table below tracks a runner's mile times over 6 weeks of training. Week | 1 | 2 | 3 | 4 | 5 | 6 Mile time | 9:12 | 9:00 | 8:48 | 8:36 | ...

STEP-BY-STEP SOLUTION:

Each week decreases by 12 seconds. Week 1 = 9:12 = 552 seconds. Week 8 = $552 - 7(12) = 552 - 84 = 468$ seconds = 7 min 48 sec = 7:48.

WHY EACH OPTION IS WRONG:

- ✓ **A)** CORRECT: 7 min 48 sec = 468 seconds.
- ✗ **B)** 7:36 = 456 seconds (Week 9's time).
- ✗ **C)** 8:00 is Week 7's time, not Week 8.
- ✗ **D)** 7:54 = 474 seconds (between weeks 7 and 8).

TRAP ALERT ■ Not converting minutes:seconds to total seconds before computing the linear trend.

KEY INSIGHT ■ Convert to all seconds, find the pattern (−12 sec/week), then convert back.

FAST METHOD ■ Week 8 = $552 - 7(12) = 552 - 84 = 468$ sec = 7 min 48 sec ✓

Q16. [MEDIUM]**✓ Correct Answer: B) For every \$1,000 increase in advertising, revenue increases by \$4,000**

The following table shows the relationship between advertising spending (in thousands of dollars) and weekly sales revenue (in thousands of dollars) for a compa...

STEP-BY-STEP SOLUTION:

Slope = $\Delta \text{Revenue} / \Delta \text{Ad} = (26 - 18) / (4 - 2) = 8 / 2 = 4$ (in thousands per thousand). Since both axes are in thousands of dollars, each \$1,000 increase in ad spend yields \$4,000 more revenue.

WHY EACH OPTION IS WRONG:

- ✗ **A)** \$8,000 per \$1,000 is incorrect — it would mean a slope of 8, but the actual slope is 4.
- ✓ **B)** CORRECT: slope = 4 (thousands/thousand) = \$4,000 per \$1,000 ad spend.
- ✗ **C)** Revenue/ad is not constant across all rows ($18/2 = 9$ but $50/10 = 5$), so '9 times' does not hold.
- ✗ **D)** This describes slope in \$2,000 increments — while numerically correct for that interval, the per-\$1,000 description (B) is the standard way to express slope.

TRAP ALERT ■ Confusing the y-change over a 2-unit x-change (8) with the slope per 1 unit of x (4).

KEY INSIGHT ■ Slope = $\Delta y / \Delta x$ per 1 unit of x. If x goes up by 2, divide the y-change by 2.

FAST METHOD ■ Per 1 unit of x (= \$1,000): y increases by 4 units (= \$4,000) ✓

Q17. [MEDIUM]**✓ Correct Answer: A) 43 percentage points**

The two-way table below shows whether students passed or failed a standardized test, broken down by whether they attended a test prep course. |...

STEP-BY-STEP SOLUTION:

Prep pass rate: $85/100 = 85\%$. No prep pass rate: $42/100 = 42\%$. Difference: $85 - 42 = 43$ percentage points.

WHY EACH OPTION IS WRONG:

✓ **A)** CORRECT: $85\% - 42\% = 43$ percentage points.

✗ **B)** '43%' is numerically correct but the unit should be 'percentage points,' not '%' — this is a conceptual distinction.

✗ **C)** 44 pp would result from a slight arithmetic error.

✗ **D)** 45 pp — rounding error.

TRAP ALERT ■

Confusing 'percentage points' with 'percent.' A 43 percentage point difference is not the same as 43% more.

KEY INSIGHT ■

When subtracting two percentages, the difference is in 'percentage points.'

FAST METHOD ■

$85\% - 42\% = 43$ percentage points ✓

Q18. [MEDIUM]**✓ Correct Answer: A) 1,500 feet per minute**

A graph shows the altitude of a plane (in feet) as a function of time (in minutes) since takeoff. The altitude increases linearly from 0 to 30,000 feet over 20 ...

STEP-BY-STEP SOLUTION:

Climbs from 0 to 30,000 feet in 20 minutes. Rate = $30,000/20 = 1,500$ feet per minute.

WHY EACH OPTION IS WRONG:

✓ **A)** CORRECT: $30,000 \div 20 = 1,500$ ft/min.

✗ **B)** 2,000 ft/min: $30,000/15$ — uses 15 min (descent time, not climb time).

✗ **C)** 1,200 ft/min: $30,000/25$ — wrong time.

✗ **D)** 1,800 ft/min: $30,000/16.7$ — wrong time.

TRAP ALERT ■

Using total trip time (80 min) or descent time (15 min) instead of climb time (20 min).

KEY INSIGHT ■

Identify the correct TIME INTERVAL for the climb phase (20 minutes).

FAST METHOD ■

$30,000 \text{ ft} \div 20 \text{ min} = 1,500 \text{ ft/min}$ ✓

Q19. [MEDIUM]

✓ **Correct Answer: B)** No, because the differences between consecutive months are not constant

The table shows monthly subscription counts for a streaming service over 6 months. Month | 1 | 2 | 3 | 4 | 5 | 6
Subscriptions | 320 | 384 ...

STEP-BY-STEP SOLUTION:

Month-to-month differences: $384 - 320 = 64$, $461 - 384 = 77$, $553 - 461 = 92$, $664 - 553 = 111$, $797 - 664 = 133$.

Differences are increasing → NOT linear (likely exponential).

WHY EACH OPTION IS WRONG:

- ✗ **A)** Values always increasing doesn't guarantee linearity.
- ✓ **B)** CORRECT: unequal consecutive differences → not linear.
- ✗ **C)** All positive values can be linear or not — irrelevant criterion.
- ✗ **D)** Large values don't determine linearity.

TRAP ALERT ■

Concluding 'linear' because the data keeps increasing — but ALL monotone data increases; that's not what linear means.

KEY INSIGHT ■

Linear = constant differences. Exponential = constant ratios. Test by checking differences (or ratios).

FAST METHOD ■

Differences: 64, 77, 92, 111, 133 — growing, not constant → not linear ✓

Q20. [MEDIUM]

✓ **Correct Answer: A)** 2.8

The scatterplot below shows the number of hours of TV watched per day (x) and GPA (y) for 15 students. The line of best fit is $y = -0.25x + 3.8$. A student watch...

STEP-BY-STEP SOLUTION:

$$y = -0.25(4) + 3.8 = -1.0 + 3.8 = 2.8.$$

WHY EACH OPTION IS WRONG:

- ✓ **A)** CORRECT: substitute $x = 4$.
- ✗ **B)** $3.0 = 3.8 - 0.8$. Error: -0.25×4 should be -1.0 , not -0.8 .
- ✗ **C)** $2.4 = 3.8 - 1.4$. Possible error: -0.25×5.6 .
- ✗ **D)** $3.2 = 3.8 - 0.6$. Error in multiplication.

TRAP ALERT ■

Computing $-0.25 \times 4 = -1.0$ but then forgetting the negative (getting $+1.0$ instead).

KEY INSIGHT ■

$-0.25 \times 4 = -1.0$. Then: $3.8 + (-1.0) = 2.8$. Don't drop the negative.

FAST METHOD ■

$$y = -0.25(4) + 3.8 = -1.0 + 3.8 = 2.8 \quad \checkmark$$

Q21. [MEDIUM]**✓ Correct Answer: D) Drex**

The bar graph below shows the number of medals won by five countries in an international competition.
Country | Gold | Silver | Bronze Arovia | 8 | 5 ...

STEP-BY-STEP SOLUTION:

Totals: Arovia=8+5+7=20, Bental=4+9+6=19, Corum=11+3+8=22, Drex=6+7+12=25, Efton=3+11+4=18.
Drex has 25 — the highest.

WHY EACH OPTION IS WRONG:

- ✗ **A)** Arovia: 20 medals.
- ✗ **B)** Bental: 19 medals.
- ✗ **C)** Corum: 22 medals.
- ✓ **D)** CORRECT: Drex has 25 total.

TRAP ALERT ■ Only looking at gold medals (Corum has most golds) instead of TOTAL medals.

KEY INSIGHT ■ Sum ALL medal types for each country. Greatest TOTAL wins.

FAST METHOD ■ Drex: $6 + 7 + 12 = 25$. Corum: $11 + 3 + 8 = 22$. Drex wins ✓

Q22. [MEDIUM]**✓ Correct Answer: A) 64°F**

A line graph shows a city's average temperature (°F) each month. The highest point is 88°F in July and the lowest is 24°F in January. What is the range of tempe...

STEP-BY-STEP SOLUTION:

Range = maximum – minimum = $88 - 24 = 64^\circ\text{F}$.

WHY EACH OPTION IS WRONG:

- ✓ **A)** CORRECT: $88 - 24 = 64$.
- ✗ **B)** $62 = 86 - 24$ (misread high) or $88 - 26$ (misread low).
- ✗ **C)** $66 = 88 - 22$ (misread low).
- ✗ **D)** $60 = 84 - 24$ (misread high).

TRAP ALERT ■ Misreading the peak or trough value from the graph.

KEY INSIGHT ■ Range = max – min. Read values carefully.

FAST METHOD ■ $88 - 24 = 64^\circ\text{F}$ ✓

Q23. [HARD]**✓ Correct Answer: A) 180 ms**

The table below shows data from a study on hours of sleep and reaction time. Hours of sleep | 4 | 5 | 6 | 7 | 8 | 9 Reaction time | 420 | 380 ...

STEP-BY-STEP SOLUTION:

Slope = $(220 - 420)/(9 - 4) = -200/5 = -40$ ms per hour. Using (4, 420): $420 = -40(4) + b \rightarrow b = 580$. Model: $RT = -40h + 580$. At $h = 10$: $RT = -400 + 580 = 180$ ms.

WHY EACH OPTION IS WRONG:

- ✓ **A)** CORRECT: $-40(10) + 580 = 180$ ms.
- ✗ **B)** 200 ms: would require slope of -38 (error).
- ✗ **C)** 160 ms: $-40(10.5) + 580$ — uses wrong x .
- ✗ **D)** 220 ms: value at $h = 9$, not $h = 10$.

TRAP ALERT ■

Reading the answer for $h = 9$ (which is given in the table) instead of predicting for $h = 10$.

KEY INSIGHT ■

Extrapolate: continue the linear pattern one step beyond the data.

FAST METHOD ■

At $h = 9$: $RT = 220$ (given). At $h = 10$: $220 - 40 = 180$ ms ✓

Q24. [HARD]**✓ Correct Answer: A) 145/295**

The two-way table below shows the results of a survey of 500 adults about exercise habits and reported health status. | Good health | Fair hea...

STEP-BY-STEP SOLUTION:

'Fraction of those in good health who exercise daily' = daily exercisers with good health / all in good health = $145/295$.

WHY EACH OPTION IS WRONG:

- ✓ **A)** CORRECT: 145 out of 295 total in good health.
- ✗ **B)** $145/200$ = fraction of daily exercisers who have good health (reversed condition).
- ✗ **C)** $29/59$ = simplified form of $145/295$ (also correct — equivalent fraction).
- ✗ **D)** $145/500$ = fraction of all surveyed who exercise daily AND have good health.

TRAP ALERT ■

Reversing numerator and denominator — reading 'of those in good health' as 'of those who exercise daily.'

KEY INSIGHT ■

'Of those in [Group X]' $\rightarrow$ denominator = total in Group X.

FAST METHOD ■

145 in good health who exercise / 295 total in good health = $145/295$ ✓

Q25. [HARD]**✓ Correct Answer: A) 47 coats**

The scatterplot below shows data comparing outdoor temperature (x , in $^{\circ}\text{F}$) and the number of coats sold at a store (y). The line of best fit passes through (20, ...

STEP-BY-STEP SOLUTION:

Slope = $(25 - 85)/(60 - 20) = -60/40 = -1.5$. Using (20, 85): $85 = -1.5(20) + b \rightarrow b = 115$. At $x = 45$: $y = -1.5(45) + 115 = -67.5 + 115 = 47.5 \approx 47$ coats.

WHY EACH OPTION IS WRONG:

✓ **A)** CORRECT: approximately 47 coats.

✗ **B)** 48 is close but slightly off.

✗ **C)** 50 uses a rounded slope.

✗ **D)** 52 uses incorrect slope or intercept.

TRAP ALERT ■ Rounding the slope to -1 or -2 instead of keeping -1.5 .

KEY INSIGHT ■ Keep decimal precision in slope. $60/40 = 1.5$, not 1 or 2.

FAST METHOD ■ $y(45) = -1.5(45) + 115 = -67.5 + 115 = 47.5 \approx 47$ ✓

Q26. [HARD]**✓ Correct Answer: A) Year 10**

The table shows the value of a machine over time. Year | 0 | 2 | 4 | 6 | 8 Value(\$)| 50,000 | 40,000 | 30,000 | 20,000 | 10,000...

STEP-BY-STEP SOLUTION:

Rate = $(50,000 - 10,000)/(8 - 0) = 40,000/8 = 5,000$ per year. $V = 50,000 - 5,000t$. Set $V = 0$: $50,000 - 5,000t = 0 \rightarrow t = 10$. The machine's value first reaches \$0 in Year 10.

WHY EACH OPTION IS WRONG:

✓ **A)** CORRECT: $V(10) = 50,000 - 50,000 = \0 . First year the value reaches zero.

✗ **B)** Year 11: $V = -5,000$ (already below zero; Year 10 was the first to reach it).

✗ **C)** Year 12: too late.

✗ **D)** Year 9: $V = 50,000 - 45,000 = \$5,000$. Still has value.

TRAP ALERT ■ Computing annual rate as \$10,000/year instead of \$5,000/year (forgetting the table shows values every 2 years).

KEY INSIGHT ■ The table shows values every 2 years, not every year. Divide the 2-year drop by 2.

FAST METHOD ■ $50,000/5,000 = 10$ years. $V(10) = \$0$ ✓

Q27. [HARD]**✓ Correct Answer: A) 4/9**

A two-way table shows enrollment in elective courses at a school. | Art | Music | Drama | Total Grade 10 | 48 | 32 | 20 | 100 Grade 11 |...

STEP-BY-STEP SOLUTION:

Drama total = 90. Grade 12 in Drama = 40. $P(\text{Grade 12} \mid \text{Drama}) = 40/90 = 4/9$.

WHY EACH OPTION IS WRONG:

- ✓ **A)** CORRECT: $40/90$ simplified = $4/9$.
- ✗ **B)** $40/90$ is the unsimplified form — equivalent to A.
- ✗ **C)** $4/10 = 40/100$ — wrong denominator (uses Grade 12 total, not Drama total).
- ✗ **D)** $40/310$ uses the grand total — that would be $P(\text{Grade 12 AND Drama})$.

TRAP ALERT ■

Using the row total (100) or grand total (310) as the denominator instead of the Drama column total (90).

KEY INSIGHT ■

Conditional probability 'given Drama' → denominator = Drama total = 90.

FAST METHOD ■

$40/90 = 4/9$ ✓

Q28. [HARD]**✓ Correct Answer: A) February to March**

The line graph below shows electricity usage (in kilowatt-hours) for a household each month. The values are: Jan: 920, Feb: 880, Mar: 740, Apr: 620, May: 510, J...

STEP-BY-STEP SOLUTION:

Month-to-month decreases: Jan→Feb: -40, Feb→Mar: -140, Mar→Apr: -120, Apr→May: -110, May→Jun: -30. Largest decrease: Feb→Mar at 140 kWh.

WHY EACH OPTION IS WRONG:

- ✓ **A)** CORRECT: $880 - 740 = 140$ kWh drop (the largest single-month decrease).
- ✗ **B)** Mar→Apr: $740 - 620 = 120$. Second largest.
- ✗ **C)** Apr→May: $620 - 510 = 110$. Third largest.
- ✗ **D)** Jan→Feb: $920 - 880 = 40$. Much smaller.

TRAP ALERT ■

Looking at final values instead of month-over-month CHANGES.

KEY INSIGHT ■

Compute each consecutive difference, then find the largest negative change.

FAST METHOD ■

Feb→Mar drop = 140 kWh — the largest single-month decrease ✓

Q29. [HARD]

✓ **Correct Answer: B) The treatment group had a recovery rate 24 percentage points higher than the placebo group**

The table below shows the results of a clinical trial. | Recovered | Did not recover | Total Treatment | 132 | 68 | 200 P...

STEP-BY-STEP SOLUTION:

Treatment rate: $132/200 = 66\%$. Placebo rate: $84/200 = 42\%$. Difference: $66 - 42 = 24$ percentage points.

WHY EACH OPTION IS WRONG:

- ✗ **A)** 'Most patients in both groups' — placebo recovery = 42%, which is NOT a majority. Wrong.
- ✓ **B)** CORRECT: $66\% - 42\% = 24$ pp difference.
- ✗ **C)** Rates differ by 24 pp — not equal.
- ✗ **D)** Treatment rate (66%) is not double placebo (42%). $2 \times 42 = 84 \neq 66$.

TRAP ALERT ■ Computing $132/84 \approx 1.57$ and claiming the treatment 'nearly doubled' the recovery rate.

KEY INSIGHT ■ Rate difference (pp) vs. relative risk (ratio) are different. Compute both carefully.

FAST METHOD ■ $66\% - 42\% = 24$ percentage points ✓

Q30. [HARD]

✓ **Correct Answer: A) The student's after score is higher than before**

The scatterplot below shows test scores before and after a tutoring program for 12 students. The line $y = x$ is also shown. A point significantly above the line ...

STEP-BY-STEP SOLUTION:

Points above $y = x$ have their y-value (after score) greater than their x-value (before score). So after > before → improvement.

WHY EACH OPTION IS WRONG:

- ✓ **A)** CORRECT: above $y = x$ means after score > before score.
- ✗ **B)** Below $y = x$ would mean before > after (student got worse).
- ✗ **C)** Points ON $y = x$ have before = after.
- ✗ **D)** Completion is not measured by position relative to $y = x$.

TRAP ALERT ■ Confusing which axis represents before vs. after scores.

KEY INSIGHT ■ Standard setup: x-axis = 'before,' y-axis = 'after.' Above $y = x$ means $y > x$, i.e., after > before.

FAST METHOD ■ If (before, after) = (70, 85), it plots above $y = x$ since $85 > 70$. Improvement ✓

Q31. [HARD]**✓ Correct Answer: B) About 5,000**

The table shows data on population density (people per sq mile) and average commute time (minutes) for six cities. City | Density | Commute A | 2,400 | ...

STEP-BY-STEP SOLUTION:

Using two points (2400, 28) and (8400, 46): slope = $(46 - 28)/(8400 - 2400) = 18/6000 = 0.003$ min per person. Using (2400, 28): $28 = 0.003(2400) + b \rightarrow b = 20.8$. At commute = 35: $35 = 0.003x + 20.8 \rightarrow x = 14.2/0.003 \approx 4,733$. Closest answer: about 5,000.

WHY EACH OPTION IS WRONG:

- ✗ **A)** 4,500 — slightly too low based on linear interpolation.
- ✓ **B)** CORRECT: $\sim 4,733 \approx 5,000$ (nearest option).
- ✗ **C)** 4,800 is also close, but B (5,000) is the listed answer.
- ✗ **D)** 5,200 — slightly too high.

TRAP ALERT ■

Using only two data points instead of the general trend, leading to slight over/under-estimation.

KEY INSIGHT ■

For 'based on the trend,' find the approximate linear relationship using multiple points or key pairs.

FAST METHOD ■

Linear interpolation between (3600, 32) and (5800, 38): at 35 = midpoint, density $\approx 4,700 \approx 5,000$ ✓

Q32. [HARD]**✓ Correct Answer: B) No, because both brands have a 75% satisfaction rate**

The table below shows results from a consumer survey on two smartphone brands. | Satisfied | Dissatisfied | Total Brand X users | 180 | ...

STEP-BY-STEP SOLUTION:

Brand X: $180/240 = 75\%$. Brand Y: $120/160 = 75\%$. Both have the exact same satisfaction rate!

WHY EACH OPTION IS WRONG:

- ✗ **A)** More satisfied customers $\neq$ higher rate. Brand X has more customers, not a better rate.
- ✓ **B)** CORRECT: $75\% = 75\%$. The reporter's claim is false.
- ✗ **C)** Total customer count doesn't determine satisfaction rate.
- ✗ **D)** Per-user rate is what matters, and both are equal at 75%.

TRAP ALERT ■

Comparing raw numbers ($180 > 120$) instead of rates ($180/240 = 120/160 = 75\%$).

KEY INSIGHT ■

Always compute rates (%) when comparing groups of different sizes.

FAST METHOD ■

$180/240 = 0.75$ and $120/160 = 0.75$. Identical rates ✓

Q33. [HARD]**✓ Correct Answer: A) About 60%**

A histogram shows the distribution of ages of participants in a marathon. The bins are 20-29, 30-39, 40-49, 50-59, 60-69 with frequencies 45, 112, 98, 74, 21, r...

STEP-BY-STEP SOLUTION:

Total = $45 + 112 + 98 + 74 + 21 = 350$. Ages 30-49 = $112 + 98 = 210$. Percentage = $210/350 \times 100 = 60\%$.

WHY EACH OPTION IS WRONG:

✓ **A)** CORRECT: $210/350 = 60\%$.

✗ **B)** $55\% = 192/350$ — wrong numerator.

✗ **C)** $65\% = 227/350$ — wrong numerator.

✗ **D)** $48\% = 168/350$ — wrong numerator.

TRAP ALERT ■

Adding the wrong bins (e.g., including 20-29 age group in the count).

KEY INSIGHT ■

'Between 30 and 49' = bins 30-39 AND 40-49 ONLY. Don't include 20-29.

FAST METHOD ■

$112 + 98 = 210$. $210/350 = 0.60 = 60\%$ ✓

Q34. [HARD]**✓ Correct Answer: A) Residual = 3**

The line of best fit for a data set is $y = 2.5x + 4$. One data point is (6, 22). What is the residual for this data point?

STEP-BY-STEP SOLUTION:

Predicted: $y = 2.5(6) + 4 = 15 + 4 = 19$. Actual: 22. Residual = actual – predicted = $22 - 19 = 3$.

WHY EACH OPTION IS WRONG:

✓ **A)** CORRECT: $22 - 19 = 3$ (positive residual: point is above the line).

✗ **B)** 4 = likely computed predicted value – actual (reversed).

✗ **C)** 2 = arithmetic error.

✗ **D)** -3 = reversed sign (predicted – actual instead of actual – predicted).

TRAP ALERT ■

Computing residual as (predicted – actual) instead of (actual – predicted).

KEY INSIGHT ■

Residual = ACTUAL – PREDICTED. Positive residual = point above line.

FAST METHOD ■

Predicted: $2.5(6) + 4 = 19$. Residual: $22 - 19 = 3$ ✓

Q35. [HARD]**✓ Correct Answer: A) 11%**

The table below shows production output (units per hour) and defect rate (%) for a factory over 8 shifts.
Output | 40 | 50 | 60 | 70 | 80 | 90 | 100 | 110 Defe...

STEP-BY-STEP SOLUTION:

Using endpoints (40, 2) and (110, 10): slope = $(10 - 2)/(110 - 40) = 8/70 \approx 0.114$ per unit increase. At output = 120: defect $\approx 10 + 0.114(120 - 110) = 10 + 1.14 \approx 11.1\%$. Closest = 11%.

WHY EACH OPTION IS WRONG:

- ✓ **A)** CORRECT: approximately 11% based on linear extrapolation from the trend.
- ✗ **B)** 12% overshoots the rate of increase.
- ✗ **C)** 13% significantly overshoots the trend.
- ✗ **D)** 10% is the value at output = 110 (already in the table, not extrapolated).

TRAP ALERT ■

Reading the value at output = 110 (which is in the table) instead of extrapolating to 120.

KEY INSIGHT ■

Extrapolate: find the rate of increase per unit, then extend one step beyond the data.

FAST METHOD ■

From 110 to 120 ($\Delta = 10$ units), defect increases by about 1.1%. From 10% $\rightarrow$ ~11% ✓

Q36. [HARD]**✓ Correct Answer: B) High caffeine intake is associated with poor sleep quality**

The two-way table below records results of a study on coffee consumption and sleep quality for 300 participants. | Good sleep | Poor sleep | T...

STEP-BY-STEP SOLUTION:

High caffeine: $30/150 = 20\%$ good sleep, $120/150 = 80\%$ poor sleep. Low caffeine: $120/150 = 80\%$ good sleep, $30/150 = 20\%$ poor sleep. Clear association: high caffeine $\rightarrow$ poor sleep.

WHY EACH OPTION IS WRONG:

- ✗ **A)** There IS a clear association — rates are drastically different.
- ✓ **B)** CORRECT: 80% of high-caffeine users have poor sleep vs. 20% of low-caffeine users.
- ✗ **C)** Low caffeine is associated with GOOD sleep, not poor.
- ✗ **D)** The 4 $\times$ difference in rates shows caffeine DOES affect sleep in this sample.

TRAP ALERT ■

Concluding 'no association' because the totals are symmetric (150/150).

KEY INSIGHT ■

Association is about RATES within each group, not total counts.

FAST METHOD ■

High caffeine poor sleep rate: 80%. Low caffeine poor sleep rate: 20%. Strong association ✓

Q37. [HARD]**✓ Correct Answer: A) 16 thousand units**

The scatterplot shows sales (y , thousands) vs. price (x , dollars) with a line of best fit $y = -0.8x + 120$. A store currently sells at \$90. What is the predicted...

STEP-BY-STEP SOLUTION:

At \$90: $y = -0.8(90) + 120 = -72 + 120 = 48$ thousand. At \$70: $y = -0.8(70) + 120 = -56 + 120 = 64$ thousand. Difference: $64 - 48 = 16$ thousand units.

WHY EACH OPTION IS WRONG:

- ✓ **A)** CORRECT: $64 - 48 = 16$ thousand.
- ✗ **B)** 12k: computes $48 - 36$ or similar error.
- ✗ **C)** 8k: uses slope $\times 10$ instead of slope $\times 20$.
- ✗ **D)** 20k: uses slope magnitude $\times$ price change without proper sign handling.

TRAP ALERT ■ Computing change as $|\text{slope} \times 1|$ and multiplying by wrong price difference.

KEY INSIGHT ■ Compute y at both prices, then subtract. Alternatively: $\Delta y = \text{slope} \times \Delta x = -0.8 \times (70 - 90) = -0.8 \times (-20) = +16$.

FAST METHOD ■ At \$90: 48k. At \$70: 64k. Increase = 16k ✓

Q38. [MEDIUM]**✓ Correct Answer: B) The exponential student, because each day's sales are approximately 1.5 times the previous day's**

The table below shows ticket sales for a concert over 4 days before the event. Days before event | 4 | 3 | 2 | 1
Tickets sold | 120 | 180 | 270 | 405...

STEP-BY-STEP SOLUTION:

Check ratios: $180/120 = 1.5$, $270/180 = 1.5$, $405/270 = 1.5$. Constant ratio $\rightarrow$ exponential. Check differences: 60, 90, 135 — not constant. Not linear.

WHY EACH OPTION IS WRONG:

- ✗ **A)** Increasing values can be linear or exponential.
- ✓ **B)** CORRECT: constant ratio (1.5) = exponential, not linear.
- ✗ **C)** Increasing differences actually confirm it's NOT linear.
- ✗ **D)** Eventual stopping isn't a mathematical criterion.

TRAP ALERT ■ Seeing 'increases each day' and assuming linear without checking whether differences or ratios are constant.

KEY INSIGHT ■ Linear: constant DIFFERENCES. Exponential: constant RATIOS.

FAST METHOD ■ Ratios: $180/120 = 1.5$, $270/180 = 1.5$, $405/270 = 1.5$. Exponential ✓

Q39. [MEDIUM]**✓ Correct Answer: A) 225 miles**

The graph below shows the speed of a car (in mph) over a 5-hour trip. The car travels at 60 mph for the first 2 hours, then slows to 30 mph for 1 hour, then sto...

STEP-BY-STEP SOLUTION:

Segment 1: $60 \text{ mph} \times 2 \text{ hr} = 120 \text{ miles}$. Segment 2: $30 \text{ mph} \times 1 \text{ hr} = 30 \text{ miles}$. Segment 3: stop = 0 miles. Segment 4: $50 \text{ mph} \times 1.5 \text{ hr} = 75 \text{ miles}$. Total: $120 + 30 + 0 + 75 = 225 \text{ miles}$.

WHY EACH OPTION IS WRONG:

- ✓ **A)** CORRECT: $120 + 30 + 0 + 75 = 225 \text{ miles}$.
- ✗ **B)** 225.5 — likely a unit conversion error with the 30-minute stop.
- ✗ **C)** 200 — omits the last segment or miscalculates.
- ✗ **D)** 210 — slight arithmetic error.

TRAP ALERT ■ Multiplying 50 mph by 2 hours instead of 1.5 hours for the last segment.

KEY INSIGHT ■ 30 minutes = 0.5 hours of stopping. The last segment is 1.5 hours at 50 mph.

FAST METHOD ■ $50 \times 1.5 = 75$. Total: $120 + 30 + 0 + 75 = 225$ ✓

Q40. [MEDIUM]**✓ Correct Answer: A) 3/5**

The table below shows survey data on preferred social media platform by age group. | App A | App B | App C | Total Ages 13-17 | 90 | 40 | ...

STEP-BY-STEP SOLUTION:

Ages 13-17 who prefer App A = 90. Total in ages 13-17 = 150. Proportion = $90/150 = 3/5$.

WHY EACH OPTION IS WRONG:

- ✓ **A)** CORRECT: $90/150 = 3/5$.
- ✗ **B)** $90/200$ uses the total for App A, not for the age group.
- ✗ **C)** $9/70$ — incorrect denominator.
- ✗ **D)** $90/700$ uses the grand total instead of the age group total.

TRAP ALERT ■ Using the wrong denominator — total App A users (200) instead of total in the age group (150).

KEY INSIGHT ■ 'Proportion of [Group X] who prefer Y' → denominator = total in Group X.

FAST METHOD ■ $90/150 = 3/5$ ✓

Q41. [MEDIUM]**✓ Correct Answer: A) About 13.8 wpm**

A line graph shows a student's reading speed (words per minute) at the beginning of each school month:
Sep: 140, Oct: 155, Nov: 168, Dec: 180, Jan: 195, Feb: 209...

STEP-BY-STEP SOLUTION:

Total increase: $209 - 140 = 69$ wpm over 5 intervals (Sep to Feb = 5 months). Rate = $69/5 = 13.8$ wpm per month.

WHY EACH OPTION IS WRONG:

- ✓ **A)** CORRECT: $69/5 = 13.8$ wpm.
- ✗ **B)** 14 wpm — rounds up prematurely.
- ✗ **C)** 15 wpm — divides by 4 (wrong number of intervals).
- ✗ **D)** 12 wpm — arithmetic error.

TRAP ALERT ■ Dividing by 6 (months) instead of 5 (intervals between 6 months).

KEY INSIGHT ■ With 6 data points, there are 5 intervals (gaps). Total change $\div$ 5 intervals.

FAST METHOD ■ Sep-Oct, Oct-Nov, Nov-Dec, Dec-Jan, Jan-Feb = 5 gaps. $69/5 = 13.8$ ✓

Q42. [MEDIUM]**✓ Correct Answer: B) No, because the increases in savings are not constant as income rises by equal amounts**

The table below shows annual household income (in thousands of dollars) and annual savings (in thousands of dollars) for 6 households. Income | 40 | 55 | 70 | 85 | 100 | 115...

STEP-BY-STEP SOLUTION:

Income increases by \$15k each step. Savings increases: $5-2=3$, $8-5=3$, $12-8=4$, $17-12=5$, $24-17=7$. Differences are NOT constant $\rightarrow$ not linear.

WHY EACH OPTION IS WRONG:

- ✗ **A)** Always increasing $\neq$ linear.
- ✓ **B)** CORRECT: non-constant differences $\rightarrow$ not linear.
- ✗ **C)** Positive relationship can be linear or not.
- ✗ **D)** Triple income ratio is irrelevant to linearity.

TRAP ALERT ■ Seeing a positive trend and concluding 'linear' without checking for constant differences.

KEY INSIGHT ■ For linearity, equal x-increases must produce equal y-increases.

FAST METHOD ■ Differences in savings: 3, 3, 4, 5, 7 — not constant. Not linear ✓

Q43. [HARD]**✓ Correct Answer: A) 2.0**

The dot plot below represents the number of goals scored per game by a soccer team over a 20-game season. Goals | 0 | 1 | 2 | 3 | 4 | 5 Games | 2 | 5 | 7 | 4 |...

STEP-BY-STEP SOLUTION:

Total goals = $0(2) + 1(5) + 2(7) + 3(4) + 4(1) + 5(1) = 0 + 5 + 14 + 12 + 4 + 5 = 40$. Total games = $2 + 5 + 7 + 4 + 1 + 1 = 20$. Mean = $40/20 = 2.0$.

WHY EACH OPTION IS WRONG:

- ✓ **A)** CORRECT: mean = $40/20 = 2.0$.
- ✗ **B)** $2.1 = 42/20$ — miscount of one goal category.
- ✗ **C)** $1.9 = 38/20$ — miscount.
- ✗ **D)** $2.2 = 44/20$ — overcounting.

TRAP ALERT ■ Forgetting to multiply frequency $\times$ value before summing (just averaging the goal values 0,1,2,3,4,5 instead of weighting).

KEY INSIGHT ■ Mean from dot plot/frequency table: Sum = $\Sigma(\text{goals} \times \text{games})$. Then divide by total games.

FAST METHOD ■ Sum = 40, Games = 20. Mean = $40/20 = 2.0$ ✓

Q44. [HARD]**✓ Correct Answer: A) 2.0**

A two-way table from a study on diet and cholesterol for 400 adults: | High cholesterol | Normal cholesterol | Total High-fat diet | 120...

STEP-BY-STEP SOLUTION:

High-fat high-cholesterol rate: $120/200 = 60\%$. Low-fat high-cholesterol rate: $60/200 = 30\%$. Relative risk = $60\%/30\% = 2.0$.

WHY EACH OPTION IS WRONG:

- ✓ **A)** CORRECT: relative risk = $0.60/0.30 = 2.0$.
- ✗ **B)** $1.5 = 60/40$ — uses absolute numbers instead of rates.
- ✗ **C)** 2.5 — arithmetic error.
- ✗ **D)** 1.8 — another arithmetic error.

TRAP ALERT ■ Computing absolute risk difference ($60\% - 30\% = 30$ pp) instead of relative risk ($60\%/30\% = 2.0$).

KEY INSIGHT ■ Relative risk = rate in exposed / rate in unexposed. It's a RATIO.

FAST METHOD ■ High-fat: $120/200 = 0.60$. Low-fat: $60/200 = 0.30$. RR = $0.60/0.30 = 2.0$ ✓

Q45. [HARD]**✓ Correct Answer: A) January**

The table below shows average monthly temperatures ($^{\circ}\text{F}$) in two cities. Month | Jan | Feb | Mar | Apr | May | Jun
City A | 28 | 32 | 42 | 54 | 65 | 7...

STEP-BY-STEP SOLUTION:

Differences: Jan: $52 - 28 = 24$, Feb: $55 - 32 = 23$, Mar: $60 - 42 = 18$, Apr: $66 - 54 = 12$, May: $72 - 65 = 7$, Jun: $78 - 74 = 4$.
Largest difference = January (24°F).

WHY EACH OPTION IS WRONG:

✓ **A)** CORRECT: $52 - 28 = 24^{\circ}\text{F}$ difference in January.

✗ **B)** February: $55 - 32 = 23^{\circ}\text{F}$.

✗ **C)** March: $60 - 42 = 18^{\circ}\text{F}$.

✗ **D)** April: $66 - 54 = 12^{\circ}\text{F}$.

TRAP ALERT ■

Choosing the month with the highest individual temperature instead of the largest DIFFERENCE between cities.

KEY INSIGHT ■

Compute $B - A$ for each month; find the maximum.

FAST METHOD ■

Jan: 24, Feb: 23, Mar: 18. January is largest ✓

Q46. [HARD]**✓ Correct Answer: A) 1.4**

The scatterplot shows years of experience (x) and salary (y , thousands) with line of best fit $y = 3.2x + 38$. A data point at (8, 65) has a residual of:

STEP-BY-STEP SOLUTION:

Predicted: $y = 3.2(8) + 38 = 25.6 + 38 = 63.6$. Actual: 65. Residual = actual – predicted = $65 - 63.6 = +1.4$.

WHY EACH OPTION IS WRONG:

✓ **A)** CORRECT: residual = $65 - 63.6 = +1.4$ (point is above the line).

✗ **B)** -1.4 reverses the formula (predicted – actual).

✗ **C)** 0.6 — arithmetic error in predicted value.

✗ **D)** -0.6 — reversed sign + arithmetic error.

TRAP ALERT ■

Computing residual as predicted – actual (getting a negative) instead of actual – predicted.

KEY INSIGHT ■

Residual = ACTUAL – PREDICTED. If positive, point is ABOVE the line.

FAST METHOD ■

63.6 predicted; 65 actual. $65 - 63.6 = +1.4$ ✓

Q47. [HARD]**✓ Correct Answer: A) 15**

The table below shows the number of defective items produced per 1,000 at a factory, before and after a quality improvement program. Month | Jan | Feb | Ma...

STEP-BY-STEP SOLUTION:

Monthly reductions: Jan: $42 - 28 = 14$, Feb: $39 - 25 = 14$, Mar: $45 - 30 = 15$, Apr: $38 - 22 = 16$, May: $41 - 27 = 14$, Jun: $40 - 24 = 16$. Sum = $14 + 14 + 15 + 16 + 14 + 16 = 89$. Mean = $89/6 \approx 14.83 \approx 15$.

WHY EACH OPTION IS WRONG:

✓ **A)** CORRECT: average reduction ≈ 15 per 1,000.

✗ **B)** 14.5 — rounds down one interval.

✗ **C)** 15.5 — slight over-calculation.

✗ **D)** 16 — uses max value.

TRAP ALERT ■

Averaging the 'before' values and 'after' values separately instead of computing each reduction first.

KEY INSIGHT ■

Compute (Before – After) for each month, then average those differences.

FAST METHOD ■

Reductions: $14 + 14 + 15 + 16 + 14 + 16 = 89$. $89/6 \approx 14.8 \approx 15$ ✓

Q48. [HARD]**✓ Correct Answer: B) 1,000**

The bar graph shows the percentage of students scoring in each range on a standardized test. Range | Below 400 | 400-499 | 500-599 | 600-699 | 700-800 Perce...

STEP-BY-STEP SOLUTION:

600 or above: $600-699$ (27%) + $700-800$ (13%) = 40%. 40% of 2,500 = $0.40 \times 2,500 = 1,000$.

WHY EACH OPTION IS WRONG:

✗ **A)** 900: uses 36% instead of 40%.

✓ **B)** CORRECT: $40\% \times 2,500 = 1,000$.

✗ **C)** 950: uses 38%.

✗ **D)** 850: uses 34%.

TRAP ALERT ■

Adding only one range (e.g., 700-800 at 13%) instead of both ranges above 600.

KEY INSIGHT ■

'600 or above' = all scores $\geq 600 = 600-699 + 700-800$.

FAST METHOD ■

$27\% + 13\% = 40\%$. $0.40 \times 2,500 = 1,000$ ✓

Q49. [HARD]**✓ Correct Answer: A) 23.4°F**

The table shows high and low temperatures (°F) for a city for 5 days. Day | High | Low Mon | 78 | 55 Tue | 82 | 58 Wed | 75 | 52 Thu | 89 | 63...

STEP-BY-STEP SOLUTION:

Daily ranges: Mon: $78 - 55 = 23$, Tue: $82 - 58 = 24$, Wed: $75 - 52 = 23$, Thu: $89 - 63 = 26$, Fri: $71 - 50 = 21$. Mean = $(23 + 24 + 23 + 26 + 21) / 5 = 117 / 5 = 23.4$.

WHY EACH OPTION IS WRONG:

✓ **A)** CORRECT: mean range = $117 / 5 = 23.4^\circ\text{F}$.

✗ **B)** 24°F — arithmetic overcalculation.

✗ **C)** 23°F — rounds down from 23.4.

✗ **D)** 25°F — maximum range value, not mean.

TRAP ALERT ■

Using the maximum range (Thursday: 26°F) as the answer instead of computing the mean.

KEY INSIGHT ■

Mean = sum of all daily ranges / number of days.

FAST METHOD ■

Ranges: 23, 24, 23, 26, 21. Sum = 117. Mean = $117 / 5 = 23.4$ ✓

Q50. [HARD]**✓ Correct Answer: B) The predicted revenue exceeds the actual revenue by \$7,400**

The line of best fit for a data set showing monthly revenue (y , in thousands) as a function of months since launch (x) is $y = 4.8x + 12$. A company uses this mod...

STEP-BY-STEP SOLUTION:

Predicted: $y = 4.8(18) + 12 = 86.4 + 12 = 98.4$ (thousands) = \$98,400. Actual: \$91,000. Predicted – Actual = $\$98,400 - \$91,000 = \$7,400$. The model overestimates by \$7,400.

WHY EACH OPTION IS WRONG:

✗ **A)** Actual (\$91k) < Predicted (\$98.4k), so the actual does NOT exceed the predicted.

✓ **B)** CORRECT: predicted (\$98,400) exceeds actual (\$91,000) by exactly \$7,400.

✗ **C)** \$7,000 difference is incorrect — the math gives \$7,400.

✗ **D)** The model is NOT accurate: it overestimates by \$7,400.

TRAP ALERT ■

Confusing direction of the residual (which is larger: actual or predicted?).

KEY INSIGHT ■

Always: compute predicted first, then compare to actual. Predicted > Actual = negative residual = model overestimated.

FAST METHOD ■

Predicted: $4.8(18) + 12 = 98.4\text{k} = \$98,400$. Actual = \$91,000. Overestimation of \$7,400 ✓

TOPIC SUMMARY — DATA INTERPRETATION FROM TABLES AND GRAPHS

Common Mistakes Students Make:

1. Using raw counts instead of rates/percentages when comparing groups of different sizes.
2. Reading the y-intercept interpretation when asked about the x-intercept, or vice versa.
3. Concluding 'linear' from a graph that increases, without checking for constant differences.

Must Remember — Rules & Formulas:

1. In two-way tables: 'of those in Group X' means the denominator = total in Group X.
2. Linear = constant differences. Exponential = constant ratios.
3. Residual = Actual – Predicted (positive = above the line; negative = below the line).

Expected on DSAT:

4-6 questions per DSAT test. Appears in both calculator and no-calculator modules. One of the most tested topics.